Supplementary material

Who Gets an ESG Score? Coverage Selection and Firm Value in Five Southeast Asian Markets

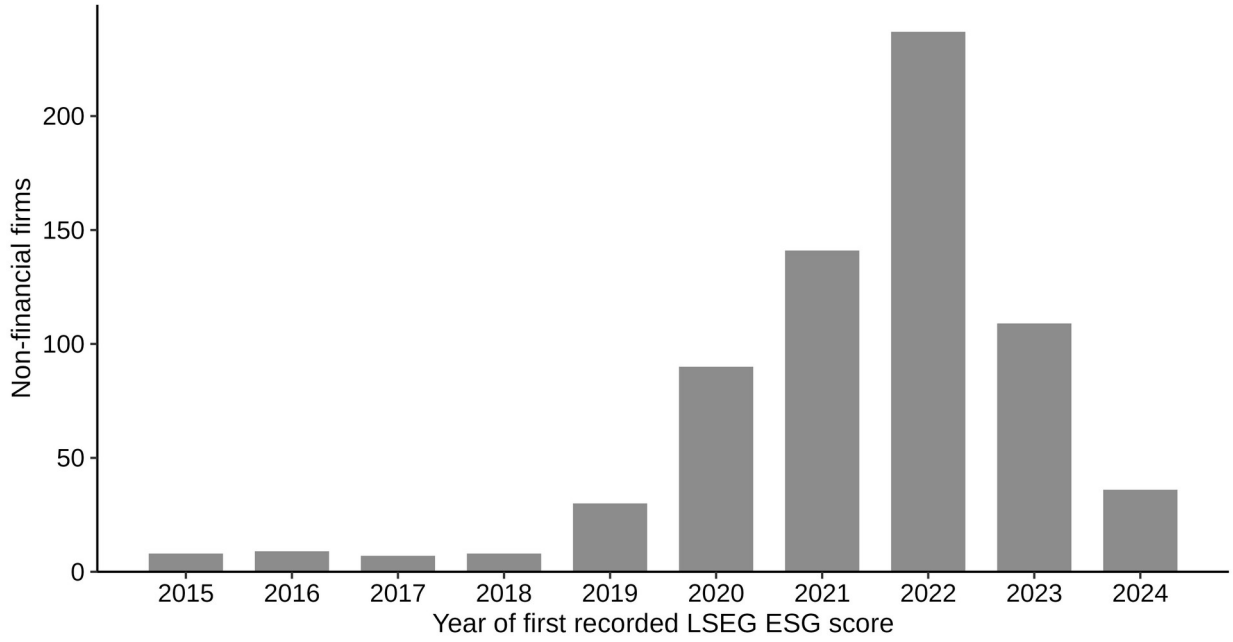
This supplementary material accompanies "Who Gets an ESG Score? Coverage Selection and Firm Value in Five Southeast Asian Markets." It presents the supplementary tables and figures first and then Appendix A, which discusses additional results. The pre-analysis plan is provided with the replication package.

Table S1. Number of non-financial firms by year of first recorded LSEG ESG score and market

First recorded score year	Indonesia	Malaysia	Philippines	Singapore	Thailand	Total
2015	2	0	2	0	4	8
2016	3	2	1	1	2	9
2017	1	4	0	0	2	7
2018	2	3	1	0	2	8
2019	1	2	0	9	18	30
2020	3	5	1	34	47	90
2021	6	96	5	3	31	141
2022	13	193	4	1	26	237
2023	8	83	0	2	16	109
2024	3	16	0	0	17	36

Note: Firms first scored in 2014 are excluded because they have no pre-coverage year.

Figure S1. Number of non-financial firms by year of first recorded LSEG ESG score



Note: Firms in the estimation sample, all five markets combined.

Table S2. Firm-years and firms available for each outcome

Outcome	Scored firm-years	Scored firms	Never-scored firm-years	Never-scored firms	Share missing
ln(MTB)	6,482	674	20,706	2,552	0.248
ln(market cap)	6,710	675	22,417	2,576	0.195
Leverage (ratio)	6,524	672	22,333	2,547	0.202
ln(assets)	6,916	675	24,814	2,611	0.123

Note: Estimation sample; the last column is the share of firm-years with a missing outcome.

Table S3. Scored firms and control firms by event year for ln(MTB)

Event year	Scored firms	Cohorts	Control firms (range)
−5	497	6	1,420–1,732
−4	521	7	1,431–1,800
−3	549	8	1,444–1,899
−2	581	9	1,455–2,065
0	615	10	1,453–2,189
1	578	9	1,442–2,034
2	483	8	1,430–1,895
3	270	7	1,418–1,796
4	141	6	1,407–1,732

Note: Scored firms are summed over cohorts; control firms show the range across cohort cells.

Table S4. Alternative specifications, inference, and initial-score splits for the average change after the first recorded LSEG ESG score

Specification	ln(MTB)	Pre-trend p	ln(market cap)	Pre-trend p
Baseline	0.002 (0.024)	0.074	−0.055* (0.030)	0.022
Baseline, market-by-industry cluster bootstrap	0.002 (0.027)	0.225	−0.055 (0.037)	0.001
R10, market-by-industry cluster bootstrap	−0.068*** (0.026)	0.237	−0.073** (0.031)	0.002
No covariates	0.025 (0.022)	0.028	0.016 (0.025)	<0.001
Size measured at g − 3	−0.008 (0.025)	0.176	−0.059* (0.032)	<0.001
Size quintile indicators	−0.010 (0.023)	0.072	−0.053* (0.029)	0.016
Scored firms within control size range	−0.021 (0.026)	0.212	−0.096*** (0.035)	0.133
Dated one year later (R10)	−0.068*** (0.024)	0.072	−0.073** (0.030)	0.002
R10, size at original g − 1	−0.074*** (0.024)	0.066	−0.060** (0.029)	0.005
Dated two years later	−0.048* (0.025)	0.023	−0.022 (0.028)	0.076
Malaysia, first recorded 2021–2022	0.021 (0.037)	0.549	−0.146*** (0.051)	0.491
Thailand and Singapore, 2019–2020	−0.009 (0.048)	0.006	0.041 (0.052)	<0.001
All other cohorts	−0.016 (0.034)	0.006	0.005 (0.037)	0.652
Cohorts first scored by 2019	0.094** (0.047)	0.003	0.185*** (0.052)	<0.001
Cohorts first scored from 2020	−0.010 (0.026)	0.199	−0.087*** (0.033)	0.106
High residual initial score	−0.012 (0.029)		−0.051 (0.035)	
Low residual initial score	0.016 (0.033)		−0.059 (0.041)	
High minus low	−0.027 (0.040)		0.008 (0.046)	

Note: Exploratory outcome-regression estimates with firm-bootstrap standard errors and the joint pre-trend p. *, **, ***: 10%, 5%, 1% (unadjusted).

Table S5. Decomposition of the gap between the static two-way fixed effects estimate and the baseline estimate

Estimator	ln(MTB)	ln(market cap)
Static TWFE, all firm-years (R6)	0.064**	0.185***
	(0.026)	(0.032)
Static TWFE, never-scored controls, event years −5 to 3	0.058**	0.149***
	(0.025)	(0.030)
Heterogeneity-robust, no covariates (R2)	0.025	0.016
	(0.022)	(0.025)
Heterogeneity-robust baseline	0.002	−0.055*
	(0.024)	(0.030)

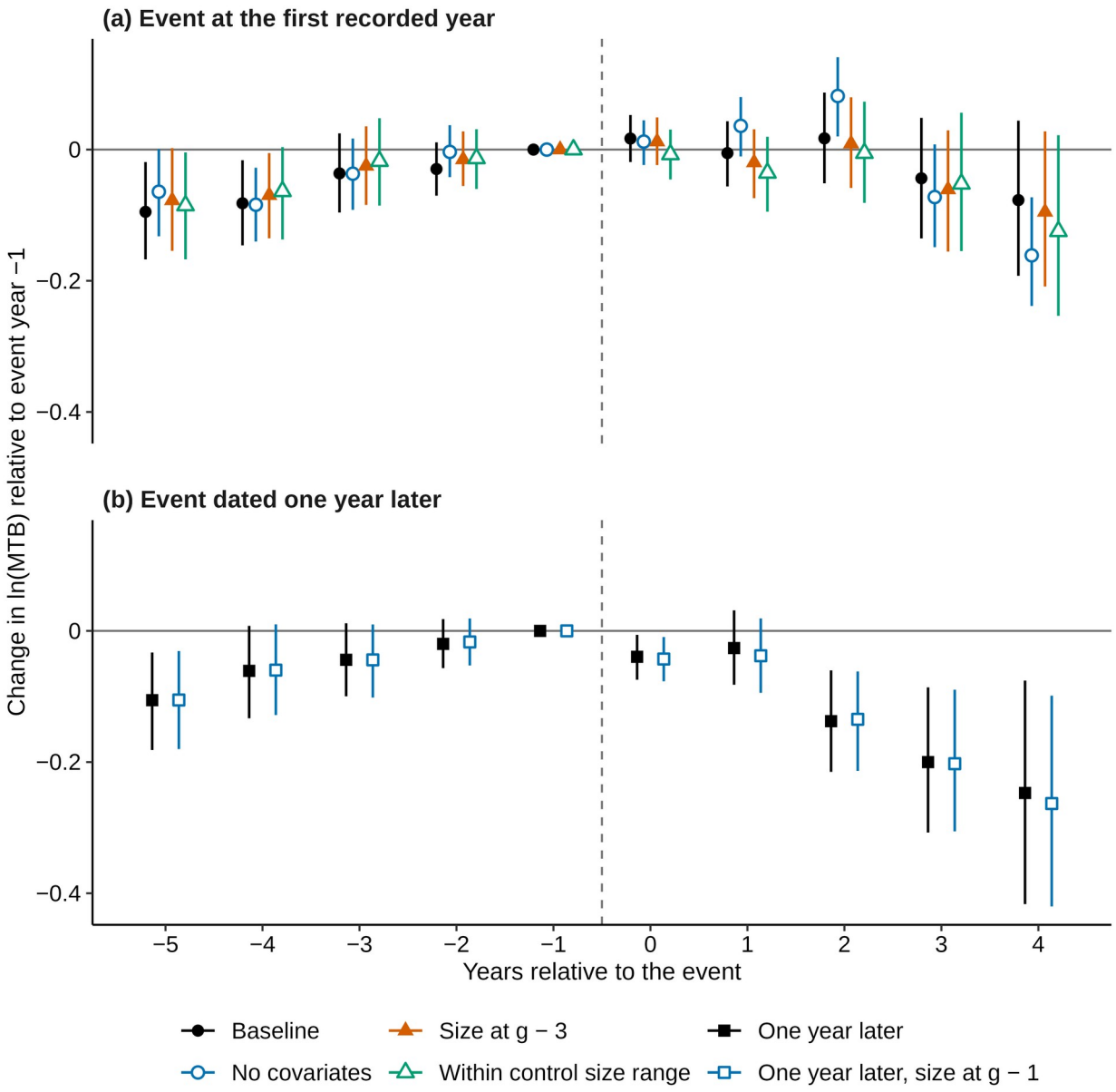
Note: Steps from Eq. (8) to the baseline Eq. (6); standard errors firm-clustered for Eq. (8), firm-bootstrap otherwise. *, **, ***: 10%, 5%, 1%.

Table S6. Average change by coverage wave under the two datings

Wave	First recorded year, ln(MTB)	One year later, ln(MTB)	First recorded year, ln(market cap)	One year later, ln(market cap)
Malaysian wave (first recorded 2021–2022)	0.021	0.012	−0.146***	−0.094*
	(0.037)	(0.033)	(0.051)	(0.052)
Thai and Singaporean wave (2019–2020)	−0.009	−0.136***	0.041	−0.038
	(0.048)	(0.046)	(0.052)	(0.053)
All other cohorts	−0.016	−0.111***	0.005	−0.076*
	(0.034)	(0.037)	(0.037)	(0.040)

Note: Outcome regression for each wave against all never-scored firms, with firm-bootstrap standard errors. *, **, ***: 10%, 5%, 1%.

Figure S2. Event-time changes in ln(MTB) under alternative specifications and dating



Note: Panel (a) varies the size covariate and support; panel (b) dates the event one year later. Firm-bootstrap 95% intervals.

Table S7. Relative-magnitude bounds for the average change after the first recorded LSEG ESG score

Outcome	$\bar{M}$	Max pre-period change	Bias bound	Robust 95% interval
ln(MTB)	0.00	0.046	0.000	[−0.046, 0.048]
ln(MTB)	0.25	0.046	0.025	[−0.071, 0.073]
ln(MTB)	0.50	0.046	0.050	[−0.096, 0.098]
ln(MTB)	1.00	0.046	0.101	[−0.147, 0.148]
ln(market cap)	0.00	0.063	0.000	[−0.109, 0.003]
ln(market cap)	0.25	0.063	0.035	[−0.144, 0.038]
ln(market cap)	0.50	0.063	0.070	[−0.179, 0.073]
ln(market cap)	1.00	0.063	0.140	[−0.249, 0.143]
Leverage (ratio)	0.00	0.012	0.000	[−0.001, 0.018]
Leverage (ratio)	0.25	0.012	0.007	[−0.007, 0.025]
Leverage (ratio)	0.50	0.012	0.013	[−0.014, 0.031]
Leverage (ratio)	1.00	0.012	0.026	[−0.027, 0.044]
ln(assets)	0.00	0.035	0.000	[0.007, 0.081]
ln(assets)	0.25	0.035	0.020	[−0.012, 0.101]
ln(assets)	0.50	0.035	0.039	[−0.032, 0.120]
ln(assets)	1.00	0.035	0.078	[−0.071, 0.159]

Note: Bias bound from Eq. (7); the robust interval widens the firm-bootstrap 95% interval by this bound.

Table S8. Firm-years with and without a market-to-book ratio

Characteristic	Scored, MTB observed	Scored, MTB missing	Never-scored, MTB observed	Never-scored, MTB missing
Firm-years	6,482	943	20,706	8,037
Median total assets (USD million)	267.9	76.4	84.1	20.8
Median market capitalization (USD million)	185.8	63.5	44.6	19.0
Mean book leverage	0.236	0.307	0.222	0.339
Share with leverage above 0.5	0.08	0.15	0.08	0.25
Share Indonesian	0.06	0.08	0.24	0.39
Share Malaysian	0.59	0.64	0.21	0.22
Mean calendar year	2019.3	2017.2	2019.5	2017.8
Share with market capitalization	1.00	0.25	1.00	0.21

Note: Estimation sample, 2014–2024; amounts in U.S. dollars after winsorization at the 1st and 99th percentiles.

Table S9. Leave-one-market-out estimates of the average change after the first recorded LSEG ESG score

Sample	ln(MTB)	ln(market cap)	Leverage (ratio)	ln(assets)	Scored / control firms
Without Indonesia	0.017 (0.024)	−0.030 (0.030)	0.011** (0.005)	0.037* (0.020)	589 / 1,734
Without Malaysia	−0.003 (0.035)	0.037 (0.038)	0.022*** (0.008)	0.046 (0.029)	256 / 1,926
Without Philippines	−0.001 (0.025)	−0.066** (0.031)	0.009* (0.005)	0.031* (0.019)	614 / 2,220
Without Singapore	−0.007 (0.025)	−0.066** (0.032)	0.005 (0.005)	0.063*** (0.019)	582 / 1,955
Without Thailand	0.004 (0.029)	−0.096** (0.037)	0.004 (0.005)	0.040* (0.021)	471 / 1,813

Note: Outcome regression with firm-bootstrap standard errors; firm counts for ln(MTB). *, **, ***: 10%, 5%, 1% (unadjusted).

Table S10. Weight of each market and coverage wave in the pooled ln(MTB) estimate

Group	Scored firm-cells	Weight in the pooled estimate
Indonesia	123	0.063
Malaysia	1,074	0.552
Philippines	52	0.027
Singapore	178	0.091
Thailand	519	0.267
Malaysian wave (first recorded 2021–2022)	858	0.441
Thai and Singaporean wave (2019–2020)	401	0.206
All other cohorts	687	0.353
Calendar year 2015	8	0.004
Calendar year 2016	16	0.008
Calendar year 2017	22	0.011
Calendar year 2018	28	0.014
Calendar year 2019	49	0.025
Calendar year 2020	126	0.065
Calendar year 2021	243	0.125
Calendar year 2022	454	0.233
Calendar year 2023	523	0.269
Calendar year 2024	477	0.245

Note: Each scored firm counts once in every post-coverage cell of the baseline ln(MTB) estimate; weights are cell shares (Eq. 6).

Table S11. Average change after the first recorded LSEG ESG score by market

Market	ln(MTB)	ln(market cap)	Leverage (ratio)	ln(assets)	Scored / control firms
Indonesia	0.054 (0.128)	−0.018 (0.109)	0.023 (0.018)	0.123* (0.070)	39 / 678
Malaysia	0.009 (0.033)	−0.114** (0.048)	0.002 (0.006)	0.033 (0.024)	372 / 486
Philippines	−0.006 (0.109)	0.101 (0.128)	0.041 (0.028)	0.025 (0.079)	14 / 179
Singapore	−0.037 (0.073)	0.143* (0.087)	0.048** (0.020)	0.013 (0.084)	46 / 452
Thailand	0.027 (0.047)	0.059 (0.059)	0.025* (0.013)	0.017 (0.040)	157 / 599

Note: Outcome regression within each market, with firm-bootstrap standard errors; firm counts for ln(MTB). *, **, ***: 10%, 5%, 1% (unadjusted).

Appendix A. Additional robustness results

A.1. Event year 4 (Section 5.2)

Only 141 scored firms from 6 cohorts reach event year 4; therefore, we plot that year in Figure 1 but exclude it from Eq. (6). Its estimates are −0.077 for MTB (p = 0.209) and 0.088 for market capitalization (p = 0.153).

A.2. Further timing variants of R10 (Section 5.4)

Under R10, the decline in log MTB loses significance once $\bar{M}$ in Eq. (7) reaches 0.23. Holding total assets at the original base year gives −0.074 (p = 0.002), so the covariate timing does not drive R10. Dating coverage two years later gives −0.048 (95% interval −0.099 to 0.000), a smaller decline that is no longer significant at the 5% level.

A.3. Reconciling the R10 and baseline event-time paths (Section 5.4, Figure S2)

The baseline path in Figure 1 does not show the R10 decline because its later event years come from early cohorts, while its first post-coverage years pool all cohorts. Re-based within each cohort to the first recorded year, the baseline point estimates for event years 3 and 4 are −0.083 and −0.167, respectively. These lie closer to the corresponding R10 estimates of −0.138 and −0.200 than the pooled baseline path suggests. We report the re-based values without standard errors, as a reconciliation of the two paths rather than a test. Table 4 of the paper carries out the full step-by-step reconciliation on a common sample.

A.4. Size covariate and support (Section 5.6, Table S4, Figure S2)

Table S4 and Figure S2 vary the size covariate and support; Table 5 of the paper reports the full set of variants. Scored-firm counts differ slightly across tables for the same estimate: Tables 4, 5, and 7 count firms in the averaged event years, while Tables 2, 3, and S7 count firms in any event-time cell. The run-up does not depend on conditioning on base-year total assets. Without covariates, the MTB run-up from event year −5 to the base year is 6.6%, weaker than the baseline (event year −5 p = 0.068; pre-trend p = 0.028), and the market capitalization run-up is 24.0%. With total assets measured two years before the base year, it is 8.0% for MTB (pre-trend p = 0.176) and 20.3% for market capitalization.

A.5. Decomposition of the TWFE gap (Section 5.8, Table S5)

Table S5 shows the decomposition of the gap between the static TWFE estimate in row R6 and the headline MTB estimate. Restricting the regression to never-scored controls and event years −5 to 3 changes the estimate to only 0.058, while the heterogeneity-robust estimator gives 0.025 without covariates and 0.002 with them. Comparing each cohort with its own base year, rather than pooling all years, accounts for approximately half of the gap. Conditioning on base-year total assets accounts for most of the rest, and the choice of controls and window for little.

A.6. Market-by-market and early versus late cohorts (Section 5.8, Table S11)

Estimated market by market in Table S11, the MTB effect is 0.009 in Malaysia, with 372 scored firms, and 0.027 in Thailand, with 157. Indonesia, the Philippines, and Singapore have few scored firms and imprecise estimates. The 58 firms first scored by 2019 that enter the MTB estimate show a positive post-coverage MTB estimate of 0.094 (p = 0.047; 95% interval 0.002 to 0.185), and the larger group first scored from 2020 shows −0.010 (p = 0.687). The early estimate rests on few firms, is one of many exploratory comparisons, and is not adjusted for multiple testing, so we do not read it as evidence of a gain for early entrants.

A.7. Breadth of the Malaysian wave (Section 5.2, Figure 2)

The Malaysian wave added 289 firms in two years, yet 565 Malaysian firms remained never-scored, so it was broad but not market-wide.

A.8. Composition of the pooled estimate (Section 5.10, Table S10)

Table S10 counts scored firm-cells, one for each scored firm in each post-coverage cell of the ln(MTB) estimate, by market and by coverage wave. These counts are the weights of Eq. (6).

A.9. Dating by wave (Section 5.4, Table S6)

Table S6 reports, for each wave, the average change with coverage dated at the first recorded score year and one year later, estimated against all never-scored firms.

A.10. Firm-years with and without an MTB ratio (Section 5.7, Table S8)

Table S8 compares scored and never-scored firm-years with and without an MTB ratio in the estimation sample. Book leverage above 0.5 and Indonesian listing are more common among firm-years without an MTB ratio.

A.11. Other robustness checks (Sections 5.5 and 5.8)

Rows R1 to R4 of Table 3 give MTB effects of −0.001 with not-yet-scored firms among the controls, 0.025 without covariates, 0.031 without the 2020 and 2021 cohorts, and 0.016 when control firms below the size range of scored firms are dropped. Under the last restriction the market capitalization effect shrinks to −0.007 and the total assets effect is 0.101, against 0.011 without covariates. Row R7 removes a linear pre-trend that passes through zero in the base year; its MTB effect is negative because a continued run-up would have raised valuations further. Row R8 adds market capitalization growth from g − 3 to the base year to the covariates; because control firms matched on past growth can themselves revert, it does not settle the reversal question. The placebo in row R5 gives 0.134 for log market capitalization ($p < 0.001$), and the static TWFE regression gives 0.185. Without Malaysia (row R9), the MTB effect is −0.003 ($p = 0.940$) and the market capitalization effect 0.037.

In the outcome regression of Eq. (3), a country or industry indicator would be unidentified in a cell with no control firm in that country or industry, and its coefficient would be set to zero. This does not occur: in all 34 cells of event years 0 to 3 for ln(MTB), and all 34 for ln(market capitalization), every indicator carried by a scored firm is also carried by a control firm.